# (Project Proposal)
# "Intelligent Document Analysis System using Computer Vision and OCR"


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## 1. Introduction

In the modern digital world, a large amount of important information still exists in the form of scanned documents, images, and printed forms. Extracting useful information from these documents manually is time-consuming, inefficient, and prone to human error.

This project proposes an **Intelligent Document Analysis System using Computer Vision and OCR (Optical Character Recognition)** that automatically processes document images and converts them into structured, machine-readable formats. The system will not only extract textual data but also analyse document layout, detect different components (text, images, tables), and organize them into a structured format.

By integrating Computer Vision techniques with OCR, the system aims to provide an efficient solution for document digitization and automated information extraction, making it highly useful in real-world applications such as banking, education, and office automation.

## 2. Problem Statement

Organizations and individuals often deal with large volumes of scanned documents, including forms, invoices, and reports. Processing these documents manually is inefficient and leads to delays and errors.

The main problems include:

- Difficulty in extracting text from scanned images
- Lack of automated document structure understanding
- Inability to separate text, images, and tables
- Time-consuming manual data entry processes

This project aims to solve these issues by developing an automated system that can intelligently analyse document images and extract structured information efficiently.

## 3. Objectives

The main objectives of this project are:

- To develop a system for preprocessing scanned document images
- To extract text using OCR techniques
- To detect and segment different document components (text, images, tables)
- To classify extracted components into meaningful categories
- To convert unstructured document data into structured formats (JSON/CSV)
- To build a scalable solution for real-world document processing

## 4. Dataset

The system will use a combination of **public datasets and custom data**.

**Selected Datasets:**

- **FUNSD (Form Understanding in Noisy Scanned Documents)**
- **RVL-CDIP Dataset (Document Classification Dataset)**
- Custom scanned documents (assignments, invoices, forms)

**Why these datasets?**

- They contain real-world document layouts
- Include structured and unstructured formats
- Suitable for OCR and layout analysis tasks

**Data Approach:**

- Pre-trained datasets will be used for understanding document structure
- Custom images will be used for testing and validation

This combination ensures both **accuracy and real-world applicability**.

## 5. Model Selection

The project will use a hybrid approach combining **pre-trained models and classical computer vision techniques**.

**Selected Models:**

**1. OCR Model:**

- Tesseract OCR
- Reason: Open-source, accurate, widely used for text extraction

**2. Document Layout Detection:**

- OpenCV (Contour Detection, Edge Detection)
- Optional: YOLO (for advanced region detection)

**3. Classification Model:**

- Support Vector Machine (SVM) or lightweight CNN
- Used to classify:
    - Text regions
    - Image regions
    - Table regions

**Approach:**

- Pre-trained models will be used for OCR
- Custom classification model may be trained for better accuracy

## 6. Methodology

The system will follow a structured pipeline:

**Step 1: Input Acquisition**

- User uploads document image or scanned file

**Step 2: Image Preprocessing**

- Convert to grayscale
- Noise removal (Gaussian Blur)
- Thresholding (binary image)
- Edge detection

**Step 3: Document Segmentation**

- Detect regions using contours
- Separate:
    - Text blocks
    - Images
    - Tables

**Step 4: Text Extraction (OCR)**

- Apply Tesseract OCR on detected text regions
- Extract raw text

**Step 5: Feature Extraction**

- Text features: word count, density
- Visual features: region size, pixel intensity

**Step 6: Classification**

- Classify regions into:
    - Text
    - Graphical elements
    - Structured components

**Step 7: Data Structuring**

- Convert extracted data into:
    - JSON
    - CSV
    - Structured text

**Step 8: Output Display**

- Show extracted content and structured format

## 7. Features of the System

The proposed system will include the following features:

- Automatic document image preprocessing
- Accurate text extraction using OCR
- Detection of document layout and structure
- Separation of text and graphical elements
- Classification of document components
- Structured data output (JSON/CSV)
- Scalable and modular architecture

## 8. Technology Stack

The system will be implemented using the following technologies:

**Programming Language:**

- Python

**Libraries and Tools:**

- OpenCV (image processing)
- Tesseract OCR (text extraction)
- NumPy (numerical operations)
- Scikit-learn (classification models)
- Matplotlib (visualization)

**Development Tools:**

- Jupyter Notebook / VS Code
- GitHub (version control and collaboration)

## 9. Team Contribution

| Team Member | Responsibilities |
|---|---|
| Member 1 | Image preprocessing and OCR integration |
| Member 2 | Document segmentation and feature extraction |
| Member 3 | Classification, structuring output, and UI handling |

Each member will also contribute to testing, debugging, and documentation.

## 10. Feasibility Analysis

**Technical Feasibility:**
The project is technically feasible as it uses well-established libraries like OpenCV and OCR tools. Pre-trained models reduce development complexity.
**Operational Feasibility:**
The system is user-friendly and requires minimal input (only document upload). It automates manual tasks effectively.
**Economic Feasibility:**
The project uses open-source tools and requires no additional cost, making it highly cost-effective.

## 11. Real-World Applications

This system can be applied in multiple domains:

- Banking systems for form and cheque processing
- Educational institutions for digitizing assignments
- Offices for document management
- Healthcare for medical record digitization
- Automated data entry systems

## 12. Conclusion

The proposed Intelligent Document Analysis System provides an effective solution for extracting and structuring information from document images. By combining Computer Vision techniques with OCR, the system can automate document processing tasks and significantly reduce manual effort.
This project is scalable, practical, and can be further enhanced into a full industry-level product by integrating advanced features such as multilingual OCR, table recognition, and web-based deployment.